Find the Error

Grade 9 · A.SSE.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the expression 5x + 12, what is the constant term? Student's answer: 5x

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **5x has the variable, so it is not constant. The constant term is 12.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the expression 7 - 2t, what is the coefficient of t? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The coefficient is the number multiplied by t, including its sign. It is -2.**

Correct answer: **-2**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A phone plan costs 25 + 10g dollars, where g is the number of GB used. What does 25 represent?
Student's answer: the cost per GB

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The cost per GB is the 10. The 25 is the fixed part that is paid no matter how much data is used.**

Correct answer: **The fixed monthly fee**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $x^2 - 9$. Student writes: $(x - 3)(x - 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$(x - 3)(x - 3) = x^2 - 6x + 9$. The correct factorization is $(x - 3)(x + 3)$.**

Correct answer: **$(x - 3)(x + 3)$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $x^2 + 5x + 6$. Student writes: $(x + 1)(x + 6)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **1 + 6 = 7, not 5. The numbers are 2 and 3, so the answer is (x + 2)(x + 3).**

Correct answer: **(x + 2)(x + 3)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $4x^2 - 25$ as a difference of squares. Student writes: $(4x - 5)(x + 5)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$4x^2 = (2x)^2$ and $25 = 5^2$. The factorization is $(2x - 5)(2x + 5)$.**

Correct answer: **$(2x - 5)(2x + 5)$**

Find the Error

Grade 9 · A.APR.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(x + 3)(x + 5)$. Student writes: $x^2 + 15$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student left out the middle terms. $(x + 3)(x + 5) = x^2 + 8x + 15$.**

Correct answer: **$x^2 + 8x + 15$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add (2x - 1) + (3x + 4). Student writes: 5x + 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The constants are -1 + 4 = 3, not 5. The sum is 5x + 3.**

Correct answer: **5x + 3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(x + 2)(x - 2)$. Student writes: $x^2 + 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The outer and inner terms cancel, leaving $x^2 - 4$. The student made a sign error.**

Correct answer: **$x^2 - 4$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an equation: a number increased by 7 is 19. Student writes: x - 7 = 19

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Increased by means add. The equation is x + 7 = 19.**

Correct answer: **x + 7 = 19**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A taxi charges $3 plus $2 per mile. The total fare is $15. Write an equation. Student writes: 2(m + 3) = 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Only the miles are multiplied by $2. The $3 is added once. The equation is 3 + 2m = 15, so m = 6.**

Correct answer: **3 + 2m = 15**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bakery sells 12 cookies per day. Write an inequality for at least 120 cookies in d days. Student writes: $12d \leq 120$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At least means greater than or equal to. The inequality is $12d \geq 120$.**

Correct answer: **$12d \geq 120$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is the point $(2, 5)$ on the graph of $y = 2x + 1$? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **2(2) + 1 = 5, so the point is on the graph.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line passes through (0, 3) and (1, 5). Write its equation. Student writes: y = 3x + 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is 2 and the y-intercept is 3. The equation is y = 2x + 3.**

Correct answer: **y = 2x + 3**

Find the Error

Grade 9 · A.CED.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the y-intercept of y = -x + 4? Student's answer: -1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the slope, -1, instead of the y-intercept. When x = 0, y = 4.**

Correct answer: 4

Find the Error

Grade 9 · A.CED.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Adult tickets cost $10 and student tickets cost $5. The total revenue is $200. Let a be the number of adult tickets and s the number of student tickets. Write an equation. Student writes: a + s = 200

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The money total uses prices. The equation is 10a + 5s = 200.**

Correct answer: **10a + 5s = 200**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A store needs at least $500 by selling x items at $12 and y items at $8. Write an inequality.
Student writes: $12x + 8y \leq 500$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At least means greater than or equal to. The inequality is $12x + 8y \geq 500$.**

Correct answer: **$12x + 8y \geq 500$**

Find the Error

Grade 9 · A.CED.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A class has 30 students. There are 6 more girls g than boys b. Write the equations. Student writes: g = b - 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 6 more girls, so g = b + 6. The total is g + b = 30.**

Correct answer: **g + b = 30 and g = b + 6**

Find the Error

Grade 9 · A.CED.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve A = lw for w. Student writes: w = Al

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied instead of dividing. Divide both sides by l: w = A/l.**

Correct answer: **w = A/l**

Find the Error

Grade 9 · A.CED.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $P = 2l + 2w$ for w. Student writes: $w = (P + 2l)/2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2l instead of subtracting. The answer is w = (P − 2l)/2.**

Correct answer: **w = (P − 2l)/2**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve d = rt for t. Student writes: t = dr

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied by r. Divide both sides by r: t = d/r.**

Correct answer: **t = d/r**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 3x + 2 = 8. A student's first step is: 3x = 10. Is this step correct? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **To isolate 3x, subtract 2 from both sides: 3x = 6. Adding 2 was the mistake.**

Correct answer: **No, it should be 3x = 6**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Explain why adding 2 to both sides of x - 2 = 5 keeps the solution the same. Student says: It changes the solution.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Doing the same operation to both sides of an equation keeps the two sides equal, so the solution stays the same.**

Correct answer: **Both sides stay equal, so the solution stays the same**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $\sqrt{x} = -2$. Student's answer: $x = 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A square root cannot be negative, so there is no solution. Squaring gives x = 4, but 4 does not work in the original equation.**

Correct answer: **No solution**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3x - 4 < 11$. Student writes: $x > 5$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Dividing by a positive number keeps the sign. $3x < 15$, so $x < 5$.**

Correct answer: **$x < 5$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $-2x + 3 \geq 9$. Student writes: $x \geq -3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Dividing by a negative reverses the sign. $-2x \geq 6$ gives $x \leq -3$.**

Correct answer: **$x \leq -3$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2(x + 4) > 2x + 8$. Student says: All real numbers are solutions.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both sides simplify to 2x + 8. The statement 2x + 8 > 2x + 8 is never true, so there is no solution.**

Correct answer: **No solution**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 = 49$. Student's answer: $x = 7$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both 7 and -7 square to 49. The solutions are x = 7 and x = -7.**

Correct answer: **x = 7 or x = -7**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $(x - 3)^2 = 16$. Student's answer: $x = 7$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found only one root. $x - 3 = 4$ or $x - 3 = -4$, so $x = 7$ or $x = -1$.**

Correct answer: **$x = 7$ or $x = -1$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 6x + 9 = 0$. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 + 6x + 9 = (x + 3)^2$, so $x = -3$. The student changed the sign.**

Correct answer: **$x = -3$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 10$ and $2x - y = 8$. Student's answer: $(4, 6)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Adding gives $3x = 18$, so $x = 6$ and $y = 4$. The student swapped the values.**

Correct answer: **$(6, 4)$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $2x + 3y = 12$ and $x - y = 1$. Student's answer: $(2, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **From $x = y + 1$, $2(y + 1) + 3y = 12$ gives $y = 2$ and $x = 3$. The student swapped the values.**

Correct answer: **$(3, 2)$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 5$ and $x + y = 7$. Student's answer: $(1, 4)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The same left side cannot equal both 5 and 7. The lines are parallel, so there is no solution.**

Correct answer: **No solution**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 2x + 1$ and $y = -x + 7$. Student's answer: (1, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2x + 1 = -x + 7$ gives $x = 2$ and $y = 5$. The student used $x = 1$ by mistake.**

Correct answer: **(2, 5)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 3x - 2$ and $y = 3x + 4$. Student's answer: $(3, 7)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both lines have slope 3 but different intercepts, so they are parallel. There is no solution.**

Correct answer: **No solution**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $y = 2x$ and $x + y = 9$ by substitution. Student's answer: (6, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x + 2x = 9$ gives $x = 3$ and $y = 6$. The student swapped the values.**

Correct answer: **(3, 6)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = 2x + 3$, what is $f(4)$? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to add 3. $f(4) = 2(4) + 3 = 11$.**

Correct answer: **11**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table lists x = 1, 2, 2 with y = 3, 4, 5. Is this a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The input 2 has outputs 4 and 5, so it is not a function.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^2$, what is $f(-3)$? Student's answer: -9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Squaring a negative number gives a positive result. $(-3)^2 = 9$.**

Correct answer: **9**

Find the Error

Grade 9 · F.IF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sequence is defined by $a(n) = 3n + 1$. What is $a(4)$? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied by 3 but did not add 1. $a(4) = 3(4) + 1 = 13$.**

Correct answer: **13**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A recursive sequence has $a(1) = 2$ and $a(n) = a(n - 1) + 3$. What is $a(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $a(2) = 5$ and $a(3) = 8$. The student added 3 to the wrong term.

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^2 - 1$, what is $f(3)$? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 1 instead of subtracting it. $f(3) = 9 - 1 = 8$.**

Correct answer: **8**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the y-intercept of y = 3x - 6? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The y-intercept is where x = 0, so y = -6. The student dropped the sign.**

Correct answer: -6

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the x-intercept of y = 2x - 8? Student's answer: -8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the slope, -1, instead of the y-intercept. When x = 0, y = 4.**

Correct answer: 4

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the maximum value of $y = -x^2 + 4$? Student's answer: 4 is the minimum.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The negative coefficient means the parabola opens down, so the vertex at (0, 4) is the maximum.**

Correct answer: **4 is the maximum**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A tank has 100 gallons and drains 4 gallons per minute. Write f(t) for the amount after t minutes.
Student writes: f(t) = 100 + 4t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The tank drains, so the amount decreases. f(t) = 100 - 4t.**

Correct answer: **f(t) = 100 - 4t**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A savings account starts with $20 and gets $5 each week. Write f(w) for the amount after w weeks. Student writes: f(w) = 5 + 20w

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The starting amount is $20, and the weekly gain is $5. f(w) = 20 + 5w.**

Correct answer: **f(w) = 20 + 5w**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sequence is 3, 6, 12, 24. Write an explicit formula for the nth term. Student writes: a(n) = 3n

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The terms double each time, so the formula is exponential. a(n) = 3 · 2^(n-1).**

Correct answer: **a(n) = 3 · 2^(n-1)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has x = 0, 1, 2 and y = 2, 4, 8. Is it linear or exponential? Student's answer: linear

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The differences are 2 and 4, but the ratio is 2 each time. The model is exponential.**

Correct answer: **Exponential**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has y = 3, 7, 11 for x = 0, 1, 2. Is it linear or exponential? Student's answer: exponential

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The difference is 4 each time, so the model is linear.**

Correct answer: **Linear**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has y = 5, 10, 20 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The ratio is 2 each time, not a constant difference. The model is exponential.**

Correct answer: **Exponential**

Find the Error

Grade 9 · F.LE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write a linear function through (1, 5) and (3, 9). Student writes: $y = 5x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is (9 − 5) ÷ (3 − 1) = 2. Then 5 = 2(1) + b, so b = 3. y = 2x + 3.**

Correct answer: **y = 2x + 3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A population of 100 grows by 50% each year. Write the exponential model. Student writes: y = 100(0.5)^t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **50% growth means the factor is 1 + 0.5 = 1.5. The model is y = 100(1.5)^t.**

Correct answer: **y = 100(1.5)^t**

Find the Error

Grade 9 · F.LE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An exponential function passes through (0, 3) and doubles each time x increases by 1. Write the function. Student writes: y = 2 · 3^x

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The starting value is 3 and the growth factor is 2. The function is y = 3 · 2^x.**

Correct answer: **y = 3 · 2^x**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the median of the data 3, 5, 5, 7, 9. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The data is already in order, and the middle value is 5. The student averaged the wrong numbers.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which display is best for categorical data, such as favorite colors? Student's answer: histogram

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Histograms show numerical data in intervals. A bar chart is best for categories.**

Correct answer: **Bar chart**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box plot has Q1 = 3 and Q3 = 9. What is the interquartile range? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the values. IQR = Q3 − Q1 = 9 − 3 = 6.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a mean of 50 and a standard deviation of 2. Group B has a mean of 55 and a standard deviation of 2. Which group has the higher center? Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The mean of Group B is 55, which is higher than 50. Group B has the higher center.**

Correct answer: **Group B**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a standard deviation of 1 and Group B has a standard deviation of 5. Which group has more spread? Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A larger standard deviation means more spread. Group B has more spread.**

Correct answer: **Group B**

Find the Error

Grade 9 · S.ID.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A data set has an outlier of 100. Which center is less affected by the outlier: mean or median?
Student's answer: mean

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The mean is pulled toward the outlier. The median is less affected because it is the middle value.**

Correct answer: **Median**